A Giant Rodent Roamed Thailand 34 Million Years Ago

From: ScienceDaily Top Environment Mined: 2026-11-12 5 min read

Imagine a rodent the size of a medium dog, scurrying through a forest 34 million years ago. That is the picture painted by two fossil teeth found in a mine in southern Thailand. The teeth belonged to a previously unknown species called Yakamys sirindhornae, and they suggest giant rodents appeared earlier and in more places than scientists had thought.

The fossils were found at the Bang Mark Pit in the Krabi **lignite** mine in Thailand's Krabi Province. The team that identified them included scientists from McGill University, Thailand's Department of Mineral Resources, and French research organizations. The remains were just two teeth: a lower second **molar** and a lower **incisor**.

To estimate the animal's size, the researchers used seven established methods that calculate body mass from tooth measurements. Their best guess: the rodent weighed about 30 kilograms, roughly 66 pounds. That would make it about twice as heavy as other known rodents from the same period, including one called Hulgana.

Estimating a long-dead animal's weight from just two teeth involves a lot of uncertainty. Virginie Millien, a biology professor at McGill University and a co-author of the study, called it 'only a **ballpark figure**.' The limited fossils also make it hard to know how the animal lived or what it ate.

The name Yakamys combines 'yak,' the Thai word for giant, with 'mys,' the Greek word for mouse or rat. The second part of the name honors Princess Maha Chakri Sirindhorn, in recognition of her support for science and **paleontological** research in Thailand.

Until now, giant rodents were thought to have evolved mostly in South America. This discovery suggests Southeast Asia also played an important role in the early history of large-bodied rodents. More fossils may one day reveal how these ancient giants lived and how they relate to the huge rodents that appeared elsewhere in the world.

Word Treasure

molar -- A molar is one of the big, flat teeth at the back of the mouth, used for chewing food.
incisor -- An incisor is one of the sharp front teeth, used for biting into food.
lignite -- Lignite is a soft, brown kind of coal that forms from plants buried long ago.
paleontological -- Paleontological means to do with the study of ancient life through fossils.
ballpark figure -- A ballpark figure is a rough guess, not an exact number.

Background you need

A fossil is the preserved remains of a plant or animal that lived long ago, often just bones or teeth that survive after everything else has gone.

Rodents are mammals with strong front teeth for gnawing, and they include animals such as mice, rats, and squirrels.

Scientists can learn a lot from a single tooth: its size and shape can hint at how big the animal was and what it might have eaten.

This discovery matters because it changes where scientists think giant rodents first evolved, pointing to Southeast Asia as well as South America.

Break it down

LEAD: Two fossil teeth from a Thai mine point to a giant rodent that lived 34 million years ago.

KEY EVENT: Scientists identify a previously unknown species, Yakamys sirindhornae, from a molar and an incisor.

+- WHERE: The Bang Mark Pit in the Krabi lignite mine, in Thailand's Krabi Province.

+- WHO: Scientists from McGill University, Thailand's Department of Mineral Resources, and French research organizations.

+- SIZE: About 30 kilograms, roughly twice as heavy as other rodents of the time, estimated with seven methods.

EVIDENCE: Only two teeth survive, so the weight is uncertain -- Millien calls it 'only a ballpark figure.'

OPEN QUESTION: How did these giants live, what did they eat, and how do they relate to giant rodents elsewhere?

Why it matters

The find rewrites the early story of giant rodents, showing Southeast Asia mattered as well as South America -- and it shows how much history can hide in just two small teeth.

Test what you remember

1. Where were the two fossil teeth found?

- (A) In a cave in Canada
- (B) In a mine in southern Thailand
- (C) On a beach in France
- (D) In a desert in South America

2. What is the new species called?

- (A) Yakamys sirindhornae
- (B) Hulgana
- (C) Millienia
- (D) Sirindhornae yak

3. About how heavy was the rodent, by the researchers' best guess?

- (A) About 30 kilograms
- (B) About 300 kilograms
- (C) About 66 kilograms
- (D) About 3 kilograms

4. What did the researchers use to estimate the rodent's size?

- (A) A complete skeleton
- (B) The animal's fur
- (C) Seven methods based on tooth measurements
- (D) Footprints in the mine

5. What did Virginie Millien call the weight estimate?

- (A) A proven fact
- (B) A wild guess with no basis
- (C) An exact measurement
- (D) Only a ballpark figure

6. What does the second part of the name honor?

- (A) The Krabi River
- (B) A famous mine
- (C) A French scientist
- (D) Princess Maha Chakri Sirindhorn

Answer key (try the quiz first!): 1: B 2: A 3: A 4: C 5: D 6: D

Questions to think about

1. The research team: The discovery shows giant rodents appeared earlier and in more places than scientists had thought, with Southeast Asia playing an important role.

2. Virginie Millien: She warns the weight is only a rough guess because so little fossil material survives.

MY THOUGHTS (at least 20 words):